

Pei Wu
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
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Academic Appointments

2024–present	PENN STATE UNIVERSITY, UNIVERSITY PARK, PENNSYLVANIA <i>Assistant Professor</i>
2023–2024	WEIZMANN INSTITUTE OF SCIENCE, REHOVOT, ISRAEL Faculty of Mathematics and Computer Science <i>Senior Postdoctoral Fellow; supervisor: Thomas Vidick</i>
Summer 2023	SIMONS INSTITUTE FOR THE THEORY OF COMPUTING, BERKELEY, CALIFORNIA <i>Research Fellow</i>
2021–2023	INSTITUTE FOR ADVANCED STUDY, PRINCETON, NEW JERSEY School of Mathematics <i>Postdoctoral Member; supervisor: Avi Wigderson</i>

Education

2015–2021	UNIVERSITY OF CALIFORNIA, LOS ANGELES <i>Ph.D., Computer Science</i> Thesis: <i>Communication and Computation</i> ; advisor: Alexander Sherstov
2013–2015	DARTMOUTH COLLEGE <i>M.S., Computer Science</i> ; thesis advisor: Amit Chakrabarti
2009–2013	NANJING UNIVERSITY, CHINA <i>B.S., Computer Science and Technology</i>

Publications

Author order is alphabetical.

- [11] ***Randomized and quantum lifting for one-way conservative NOF model***
H. Wang, **P. Wu**
The 51st International Symposium on Mathematical Foundations of Computer Science (MFCS 2026)
- [10] ***Quantum algorithms on edge lists: hiding, shuffling, and cycle finding***
A. S. Gilani, D. Wang, **P. Wu**, X. Zhou
The 53rd International Colloquium on Automata, Languages, and Programming (ICALP 2026). *Also accepted as a contributed talk at TQC 2026*
- [9] ***Quantum Merlin–Arthur with an internally separable proof***
R. Bassirian, B. Fefferman, I. Leigh, K. Marwaha, **P. Wu**
The 21st Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC 2026), contributed talk
- [8] ***Coherence in property testing: quantum-classical collapses and separations***
F. G. Jeronimo, N. Magrafta, J. Slote, **P. Wu**
The 28th Annual Quantum Information Processing Conference (QIP 2025)
- [7] ***Dimension-independent disentanglers from unentanglement and applications***
F. G. Jeronimo, **P. Wu**
The 39th Computational Complexity Conference (CCC 2024)

- [6] *Pseudorandom and pseudoentangled states from subset states*
F. G. Jeronimo, N. Magrafta, **P. Wu**
The 19th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC 2024), poster
- [5] *An optimal “it ain’t over till it’s over” theorem*
R. Eldan, A. Wigderson, **P. Wu**
The 55th ACM Symposium on Theory of Computing (STOC 2023)
- [4] *The power of unentangled quantum proofs with non-negative amplitudes*
F. G. Jeronimo, **P. Wu**
The 55th ACM Symposium on Theory of Computing (STOC 2023)
- [3] *An optimal separation of randomized and quantum query complexity*
A. A. Sherstov, A. A. Storozhenko, **P. Wu**
The 53rd ACM Symposium on Theory of Computing (STOC 2021). *SIAM Journal on Computing*, 52(2):525–567, 2023
- [2] *Near-optimal lower bounds on the threshold degree and sign-rank of AC^0*
A. A. Sherstov, **P. Wu**
The 51st ACM Symposium on Theory of Computing (STOC 2019). *Invited to appear in SIAM Journal on Computing (special issue for STOC 2019)*
- [1] *Optimal interactive coding for insertions, deletions, and substitutions*
A. A. Sherstov, **P. Wu**
The 58th Annual Symposium on Foundations of Computer Science (FOCS 2017). *IEEE Transactions on Information Theory*, 65(10):5971–6000, 2019

Survey and Expository Article

The QMA(2) universe: complexity, entanglement, and optimization
F. G. Jeronimo, I. Leigh, **P. Wu**
Invited survey, ACM SIGACT News, 57(1):64–99, 2026

Manuscripts and Preprints

Optimal quantum de Finetti theorems via argmax rounding
F. G. Jeronimo, **P. Wu**, H. Xu

An argmax principle for sum-of-squares relaxations on the sphere
F. G. Jeronimo, **P. Wu**, H. Xu

An optimal analysis of the product test
J. Bekey, F. G. Jeronimo, **P. Wu**

The power of unentanglement without destructive interference
Y. Liu, **P. Wu**

Advising

HAOYU WANG
Ph.D. student (in progress), Penn State University

HAOCHEN XU
Ph.D. student (in progress), Penn State University

JOHN BRENTON
Undergraduate researcher, Penn State University

Teaching

Fall 2026	CMPSC 360 Discrete Mathematics
Spring 2026	CMPSC 564 Complexity of Combinatorial Problems
Fall 2025	CMPSC 464 Theory of Computation

Selected Talks and Seminars

Aug–Sep 2024	<i>“Coherence in property testing: quantum-classical separations and collapses”</i> Center on Frontiers of Computing Studies, Peking University; Theory Seminar, Penn State University
Mar 2024	<i>“Multiple Merlins in the modern era”</i> Theory Lunch, Weizmann Institute of Science
Apr 2023–Jan 2024	<i>“The power of unentangled proofs with non-negative amplitudes”</i> Simons Institute; Weizmann Institute of Science; The University of Texas at Austin; Nanjing University; QIP 2024
Jun 2023	<i>“It ain’t over till it’s over”</i> “Analysis and TCS,” Simons Institute
Sep–Nov 2022	<i>“Random restrictions on Boolean functions with small influences”</i> Princeton University; Shandong University; Nanjing University; DIMACS and Rutgers University
Nov 2022	<i>“Polynomial method in communication complexity”</i> CS/DM Seminar, Institute for Advanced Study
Oct 2021	<i>“Recent progress on query complexity”</i> Two lectures, CS/DM Seminar, Institute for Advanced Study
Apr 2021	<i>“Optimal separation of randomized and quantum query complexity”</i> Algorithm and Complexity Seminar, University of Waterloo (online)
Feb 2020	<i>“Settling the threshold degree and sign-rank of AC^0”</i> Invited plenary talk, Southern California Theory Day, University of California, Riverside

University Service

2025–2026	DEPARTMENT HIRING COMMITTEE
2025	COMPUTER SCIENCE AND ENGINEERING SEMINAR SERIES Organizer
2025	GRADUATE QUALIFYING COMMITTEE

Professional Service

Program committee: FSTTCS 2026

Conference reviewing: STOC/FOCS, CCC, ITCS, ICALP, STACS, QIP, TQC

Journal referee: PRX, SICOMP, TIT, *Quantum*, *Algorithmica*, JCSS